MILITARY SPECIFICATION

HEATERS, LUBRICATING OIL, ELECTRIC, NAVAL SHIPBOARD

1. SCOPE

1.1 Scope. - This specification covers heaters which are stationary, self-contained, and electrically operated, and are used for heating lubricating oil for purification or warm-up.

1.2 Classification. - Lubricating oil heaters shall be of the following types as specified (see 6.2):

Type A - Immersion heater.
Type B - Circulation heater.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-584 - Bolts, Finned Neck; Key Head: Machine; Ribbed Neck; Square Neck; Tee Head.
QQ-C-591 - Copper-Silicon, Copper-Zinc-Silicon, and Copper-Nickel-Silicon Alloys: Rod, Wire, Shapes, Forgings, and Flat Products, (Flat Wire, Strip, Sheet, Bar, and Plate).
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought.
QQ-S-691 - Steel Plate, Carbon-Silicon, Carbon Molybdenum and Manganese-Molybdenum Alloys Hot Rolled, (Marine Boiler Quality).

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-B-857 - Bolts, Nuts, and Studs.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems. Requirements For.
MIL-D-1000/2 - Drawings, Engineering and Associated Lists.
MIL-C-2174 - Controllers, Direct-Current, Naval Shipboard.
MIL-C-2212 - Controllers, Alternating-Current, Naval Shipboard.
MIL-P-5516 - Packing, Preformed, Petroleum Hydraulic Fluid Resistant, 160°F.
MIL-P-15024 - Plates, Identification—Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-C-15726 - Copper-Nickel Alloy, Rod, Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate) and Forgings).
MIL-T-16420 - Tube, 70-30 and 90-10 Copper-Nickel Alloy, Seamless and Welded.
MIL-L-17331 - Lubricating Oil, Steam Turbine (Noncorrosive).
MIL-A-17472 - Asbestos Sheet, Compressed (Gasket Material).
MIL-F-20042 - Flanges, Pipe, Bronze (Silver Brazing).
MIL-T-20157 - Tube and Pipe, Carbon Steel, Seamless.
MIL-G-21032 - Gaskets, Metallic-Asbestos, Spiral Wound (For ASA Commercial Flanged Joints) in Piping Systems.
MIL-H-22577 - Heating Elements, Electrical; Cartridge, Strip and Tubular Type.
MIL-S-22698 - Steel Plate; Carbon, Structural, For Ships.

STANDARDS

FEDERAL

FED-STD-791 - Lubricants, Liquid Fuels, and Related Products; Methods of Testing.

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MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-278 - Fabrication, Welding and Inspection of Machinery, Piping and Pressure Vessels for Ships of the United States Navy.
MIL-STD-769 - Thermal Insulation Requirements for Machinery and Piping.

DRAWINGS

BUREAU OF SHIPS

9000-S6202-73907 - Light, Indicator, Switches, 2-Lamp SPF, types B-27A thru B-27G.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N.Y. 10016.)

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20401.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D974 - Method of Test for Neutralization Number by Color-Indicator Titration.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

UNITED STATES OF AMERICA STANDARDS INSTITUTE (USAS)

B16.5 - Steel Pipe Flanges and Flanged Fittings.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, N.Y. 10016.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials. - Unless otherwise specified herein, or in the contract or order, materials shall be in accordance with tables I and II.

Table I - Ferrous materials.

Part	Material	Applicable document
Shell and pipe connections	Tubing, seamless drawn carbon steel	MIL-T-20157
	or	
	Plate, welded, carbon-silicon steel, marine boiler quality	QQ-S-691, class B

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Table I - Ferrous materials-(Cont'd)

Part	Material	Applicable document
Head and mounting plate	Plate, carbon-silicon steel, marine boiler quality	QQ-S-691, class B
Flanges for shell and pipe connections	Steel, carbon	USAS B16.5
Plates and mounting lugs	Plate, carbon-silicon steel, marine boiler quality Plate, steel, carbon, structural	QQ-S-691, class B MIL-S-22698
Stay rods, plugs, bolts, nuts and studs	Steel, carbon	MIL-B-857, grade 1 or FF-B-584

Table II - Nonferrous materials.

Part	Material	Applicable document
Shell and pipe connections	Copper-nickel alloy, composition 90-10 or Copper-nickel alloy, composition 70-30	MIL-T-16420 MIL-C-15726
Head and mounting plate	Copper-nickel alloy	MIL-C-15726
Flanges for shell and pipe connections	Bronze or Copper-nickel alloy	MIL-F-20042 MIL-C-15726
Plates and mounting lugs	Copper-nickel alloy	MIL-C-15726
Stay rods	Nickel-copper-aluminum alloy	QQ-N-286, class B
Bolts, nuts, studs and plugs	Nickel-copper alloy or Copper silicon	MIL-B-857 QQ-C-591

3.1.1 Gaskets.- The following materials shall be used for gaskets, as applicable:

(a) Asbestos sheet in accordance with MIL-A-17472.
(b) Metallic-asbestos, spiral wound, in accordance with MIL-G-21032.
(c) O-rings, class A, in accordance with MIL-P-5516.

3.2 Design.-

3.2.1 Heating unit assembly.- The heating unit assembly shall consist of a circular mounting plate or flange, through which U or L shaped heater elements are inserted. The elements shall be welded or brazed to the plate at each penetration. The other end of the elements shall have spacers or be braced between elements for rigidity, and to provide a means of support to the heater case or sump tank. The design shall permit easy withdrawal from the sump tank or heater shell, without the removal of connecting piping.

3.2.2 Heater elements.- Heater elements shall be of the tubular type conforming to MIL-H-22577. The terminal nuts and bolts shall be of nickel-copper alloy in accordance with MIL-B-857, or AISI 303 stainless steel. Each heating element assembly shall be provided with a connection wiring diagram glued to the underside of the terminal cover plate.

3.2.3 Terminal box.- The electrical connections to the heater elements shall be enclosed in a water-tight terminal box mounted on the flange, and provided with a removable cover. Holes for cable connections to the terminal box shall be provided as specified (see 6.2).

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3.2.4 Mounting flange. - The mounting flange shall be marked to insure installation of the heating unit assembly in only one position. In order to insure interchangeability, the tolerance for the location of the drilled flange holes shall be within .010 inch of the true design location.

3.2.5 Thermostatic switches. - Heaters shall be provided with thermostatic operating switches of an adjustable immersed bulb type. The switch contacts shall open within plus or minus 10°F. of the thermostat setting within the temperature range. The switch contacts shall close before the temperature of the oil has dropped to 90 percent of the thermostat temperature setting. Switches for alternating current (a.c.) circuits shall be suitable for operation on 115 volts. Switches for direct current (d.c.) circuits shall be suitable for operation on 115 and 230 volts. Thermostatic switches shall conform to MIL-C-2174 or MIL-C-2212 as applicable. The maximum temperature of the oil in any part of the heater at the time the thermostatic switch contacts open, shall not exceed the thermostat temperature setting by more than 15°F.

3.2.6 Contactor. - When specified (see 6.2), a magnetic contactor in a dripproof case for bulkhead mounting shall be furnished with each heater. The contactor shall conform to MIL-C-2212 or MIL-C-2174, as applicable. An "on-off" switch, operating in the control circuits, shall be mounted with each contactor. The control circuit shall be arranged such that manual reset will be necessary after the contactor opens due to opening of the safety switch, or from overload. An indicator light in accordance with Drawing 9000-S6202-73907 shall be provided at the "on-off" switch to indicate that the heater element is energized.

3.2.7 Life. - The heater assembly and thermostatic switches shall be designed for 10,000 hours of trouble free operation, and 50,000 "on-off" cycles at normal operating conditions.

3.2.8 Voltage. - Heaters shall be suitable for operation on one of the following power supplies as specified (see 6.2):

(a) 440 volts (v.), 3 phase (ph.), 60 cycle (c.), a.c.
(b) 115 v. or 230 v., d.c.
(c) Special voltages.

3.2.9 Dielectric strength. - The heaters shall satisfactorily withstand the dielectric strength test specified in 4.2.4.

3.2.10 Shock. - Heaters shall be capable of withstanding the high-impact (H.I.) shock tests of 4.2.6 for grade A, class I, equipment as defined in MIL-S-901.

3.2.11 Vibration. - Heaters shall be capable of withstanding the vibration test of 4.2.7 without mechanical damage or malfunctioning.

3.2.12 Screw thread standards. - Screw threads shall be in accordance with Handbook H28. The design shall utilize standard tools to the greatest extent possible. Standard tools are defined as those tools listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Government Inspector).

3.2.13 Welding and allied processes. - Welding and allied processes shall be in accordance with MIL-STD-278.

3.3 Type A heaters. -

3.3.1 Heater unit. - Unless otherwise specified (see 6.2), the type A heater unit shall be designed for installation in a lubricating oil tank in a horizontal position. When vertical installation is required through the tank top, the elements shall be designed so that the heating position is at the bottom of the tank.

3.3.2 Safety switch or liquid level control. - A safety switch or a liquid level controll, as specified (see 6.2), shall be provided to deenergize the heater if the elements are not completely immersed in oil. The switch contacts shall open when the sheath temperature of the elements reaches 450°F. ± 50°F. When specified (see 6.2), the safety switches shall have a special setting.

3.3.3 Watt density. - The maximum watt density for type A heaters shall be 16 watts per square inch of heating element.

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3.3.4 Capacity. - The type A heater shall be capable of heating a specified volume of lubricating oil. military symbol 2190TEP conforming to MIL-L-17331 through a specified temperature rise within a specified time (see 6.2).

3.4 Type B heaters. -

3.4.1 Heater unit. - The type B heater unit shall be designed so that the flow of oil over all heating surfaces is evenly distributed to prevent localized hot spots. Hot spots are considered to be areas where oil temperatures may exceed outlet temperatures by as much as 40°F.

3.4.2 Shell. - The heater assembly shall be enclosed in a shell having inlet and outlet connections suitable for connecting into a pressure lubricating oil circulating system.

3.4.3 Pipe connections. - Flanged inlet and outlet connections shall be welded to the shell. The inlet connection shall be at the bottom of the shell, and the outlet at the top for self venting of the heater. In addition, straight threaded connections shall be provided for draining the heater, and for continuous venting. These connections shall be closed by plugs with O-ring gaskets (see 3.1.1). Steel flanges shall be in accordance with USAS B16.5, 150 p.s.i. Nonferrous flanges shall meet the requirements of MIL-F-20042.

3.4.4 Mounting. - Heaters may have mounting lugs welded to the heater shell. The lugs shall be drilled and tapped, for bolting the heater to the ship structure. The lugs shall be flush with the insulation and lagging if provided. Other means of mounting to the ship structure, such as clamp bands are acceptable.

3.4.5 Insulation. - When specified (see 6.2) heaters shall be insulated and lagged in accordance with MIL-STD-769. The lagging shall not interfere with removal of the heater from the shell.

3.4.6 Watt density. - The watt density shall not exceed 20 watts per square inch of heating element.

3.4.7 Capacity. - The heater shall be capable of heating military symbol oil 2190TEP conforming to MIL-L-17331 at the rate flow specified to a minimum temperature rise of 120°F. (see 6.2).

3.4.8 Pressures. - The heater shall be designed for operation at 100 pounds per square inch (p.s.i.) pressure, and shall be capable of withstanding a hydrostatic test of 150 p.s.i. The allowable pressure drop through the heater when operating at rated flow and 130°F. outlet temperature, shall not exceed 5 p.s.i.

3.5 Repair parts. -

3.5.1 Onboard repair parts. - Onboard repair parts shall be furnished in accordance with table III for each type and size of heater.

Table III - Onboard repair parts.

Name of part	Quantity based on installed heaters		
	1 to 5	6 to 10	11 to 20
Heater element assembly	1	1	2
Thermostatic elements	1	2	4
Switch assembly	1	2	3
Gaskets and seals	100 percent	100 percent	100 percent
Controller repair parts in accordance with MIL-C-2174 or MIL-C-2212	As applicable		

3.5.2 Complete components and all serviceable parts including onboard repair parts of 3.5.1 shall be listed in accordance with MIL-P-15137.

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3.6 Drawings. - Drawings shall be types II and III in accordance with MIL-D-1000/2. Drawings shall be approved by the command or agency concerned. In addition to the data required by MIL-D-1000/2, drawings shall include the following:

(a) Type II drawings:
  (1) Cross-sectional and longitudinal view of the heater assembly with details of construction (type B only).
  (2) Dimensioned outline of heater assembly.
  (3) Wiring diagram.
  (4) Operating electrical characteristics.
  (5) Number of heating elements; wattage of each and total watts.
  (6) Rate of oil flow gallons per hour (type B only).
  (7) Temperature rise (type B only).
  (8) Weight, including repair parts.

(b) Type III drawings:
  (1) Capacity in gallons per hour (type B only).
  (2) Temperature rise (°F.) (type B only).
  (3) Number of heating elements including the watts of each element and total watts.
  (4) Electrical characteristics.
  (5) Oil pressure drop, p.s.i. at rated flow (type B heaters).
  (6) Hydrostatic test pressure.
  (7) Reference to thermostatic switch and contactor drawings.

3.6.1 Drawings of thermostatic switches and magnetic contactors shall be furnished in accordance with the applicable component specification.

3.7 Manuals. - Manuals shall be type III in accordance with MIL-M-15071 in the quantities specified (see 6.2).

3.8 Identification and information plates. - Each heater shall have an identification plate in accordance with types A, B, or C of MIL-P-15024, mounted on the terminal box or on the heater lagging. The plate shall contain the following data:

(a) Lubricating oil heater.
(b) Capacity in gallons per minute (type B only).
(c) Serial number, and type of heater.
(d) Electrical characteristics, volts, wattage, phases, frequency and number of elements.
(e) Federal stock number (if available).
(f) Name and address of manufacturer.

3.9 Workmanship. - Heaters shall be free from defects affecting their appearance or that may affect their operation. Brazed penetration joints for heater mounting plate shall evidence a high degree of penetration of alloy and are not to be cold jointed. Mating surfaces of flanges and mounting pads shall be machined true to provide interchangeability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First unit tests. - One heater of each type and capacity offered for delivery on the contract or order shall be subjected to the first unit tests specified in 4.2.1 through 4.2.7, as applicable.

4.2.1 If the command or agency concerned is satisfied that the heaters conform to this specification, a waiver may be granted to omit first unit test on subsequent contracts or orders. A first unit test shall be made after any change in design which affects performance characteristics. If production test data reveal variations beyond the normal manufacturing tolerance, a complete first unit test shall be made on that particular type or size of heater to demonstrate that it conforms to this specification.

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4.2.2 Hydrostatic test. - The heater shall be tested at a hydrostatic pressure of 150 pounds per square inch gage (p.s.i.g.).

4.2.3 Calibration. - The calibration of the thermostatic switch shall be checked by a controlled temperature liquid bath system, or by thermocouples attached to the sensing bulb of the temperature control to indicate the temperature at which the switch contacts opens at various settings within the specified range.

4.2.4 Dielectric test. -

4.2.4.1 General. - The dielectric test shall be made after all other tests have been completed. The test voltage shall be equal to twice the rated value of the terminal voltage plus 1000 volts. The frequency shall be not less than 60 c., and the wave shape shall approximate a sine wave. The test voltage shall be applied continuously for a period of not less than 1 minute.

4.2.4.2 Measurements of test voltage. - The measurements of the voltage used in the dielectric test shall be made by means of a voltmeter whereby the meter derives its voltage directly from the high voltage circuit, either by means of a voltmeter coil placed in the testing transformer or through an auxiliary potential transformer.

4.2.4.3 Points of application. - The test voltage shall be successively applied between each electric circuit and the frame with all other electric circuits and metal parts grounded.

4.2.5 Performance test for type B heater. - Type B heater and associated controls shall be operated under normal conditions as specified in the contract or order. With inlet oil temperatures and flow rates as given in table IV, record the following:

(a) Outlet temperature (type B only).
(b) Volts and amperes.
(c) Pressure drop from inlet to outlet of heater (type B only).
(d) Number of heater elements energized.
(e) Thermocouple readings for type B only, (Thermocouples are used to indicate temperatures in areas likely to be hot spots within the heater due to insufficient oil flow.).

Table IV - Temperatures and flow rates.

Flow g.p.m.	25 percent capacity			50 percent capacity			75 percent capacity			100 percent capacity		
Inlet oil temperatures °F.	40°	60°	90°	40°	60°	90°	40°	60°	90°	40°	60°	90°
Outlet temperatures °F.												
Volts												
Amperes												
Pressure drop												
Elements energized												
Thermocouple no. 1												
Thermocouple no. 2												

4.2.5.1 Venting. - The venting capability of the type B heater shall be tested under the following conditions:

(a) Listing 15 degrees to port; listing 25 degrees to starboard.
(b) Trimmed 30 degrees forward end up and 30 degrees forward end down.

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Each run will be for a 1 hour period with inlet oil at 90°F. and rated flow. During the test air will be introduced into the oil inlet to determine the venting ability of the heater under adverse conditions. Open the vent connection periodically and check for presence of air to ensure that the heater is venting properly. Monitor areas of likely hot spots by thermocouples and record the temperatures. At the completion of the test visually examine the elements for damage and traces of carbon deposits due to the existence of an unvented air procket.

4.2.5.2 Temperature control. -With the type B heater completely filled with oil and no circulation, apply full power for 8 hours. Set thermostatic switch to cut in at 155°F. decreasing transient and cut out at 160°F. increasing transient. Monitor areas of possible hot spots and record temperatures at 1 hour intervals. At the conclusion of the test, subject a sample of oil to the following tests:

(a) Precipitation number (maximum 0.01) method 3101.5 of FED-STD-791.
(b) Neutrality, qualitative (maximum 0.20) method 5101 of FED-STD-791.
(c) Neutralization number as specified in ASTM D974.

4.2.6 Shock test. - The heater shall be subjected to the type A, H.I. shock test of MIL-S-901 for grade A, class I equipment.

4.2.6.1 Disposal of shock tested heaters. - The following shall apply to shock tested heaters:

(a) Heaters which have been subjected to the H.I. shock test and have failed to meet the requirements of this specification shall not be offered for delivery, either in whole or any of their parts.
(b) Heaters which have been subjected to the H.I. shock test and have successfully passed this test shall be retested in accordance with 4.2.2, 4.2.3 and 4.2.4 prior to delivery.

4.2.7 Vibration test. - The heater shall be subjected to the type I vibration test of MIL-STD-167.

4.3 Sampling for production inspection. -

4.3.1 Lot. - All heaters of the same type and capacity offered for delivery at one time shall be considered a lot.

4.3.2 Sampling for examination. - A random sample of heaters shall be selected from each lot in accordance with MIL-STD-105 at inspection level II for the examination of 4.4.1. The acceptable quality level (AQL) shall be 1.5 percent defective.

4.3.3 Sampling for production tests. - A random sample of heaters shall be selected from each lot in accordance with MIL-STD-105 at inspection level II for the tests specified in 4.5. The AQL shall be 1.5 percent defective.

4.4 Examination. -

4.4.1 Heaters. - Each heater selected in accordance with 4.3.2 shall be thoroughly examined to ascertain that the material, workmanship, and design are in conformance with this specification.

4.4.2 Repair parts. - All repair parts shall be carefully examined to determine that the materials, workmanship, and finish are first class in every respect and that they are exact duplicates of those used in the heaters.

4.5 Tests. -

4.5.1 Each heater selected in accordance with 4.3.3 shall be subjected to the tests specified in 4.2.3 and 4.2.4.

4.5.2 Each heater furnished under the contract or order shall be tested in accordance with 4.2.2.

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5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Heaters. -

5.1.1.1 Preservation and packaging. - Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. -

5.1.2.1 Packaging. - Parts such as thermostatic elements, switch assemblies and so forth shall be unit protected and packaged in accordance with method III of MIL-P-116. Gaskets shall be packaged in accordance with method IA, submethod IA-8 of MIL-P-116. Gaskets shall be packaged flat and provided with fiberboard stiffeners to prevent deformation. Parts shall be individually packaged except when used in sets.

5.1.2.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.2.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging. -

5.2.1.1.1 Level A. - The heaters shall be preserved and individually packaged by method III of MIL-P-116. All openings to the heater shall be sealed to prevent entrance of foreign material.

5.2.1.2 Packing. -

5.2.1.2.1 Level A. - Heaters, packaged as specified, shall be individually packed in boxes conforming to any one of the following specifications at the option of the supplier:

Specification	Type or class
PPP-B-601	Overseas type
PPP-B-621	Class 2

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5.2.1.2.2 Level B. - Heaters, packaged as specified, shall be individually packed in boxes conforming to any one of the following specifications at the option of the supplier:

Specification	Type or class
PPP-B-601	Domestic type
PPP-B-621	Class 1

5.2.1.2.3 Boxes shall be modified by the addition of skids in accordance with the applicable box specifications. Cushioning, anchoring, blocking or bracing shall be in accordance with MIL-STD-1186. Box closure shall be as specified in the applicable box specification or appendix thereto.

5.2.1.3 Marking. - In addition to any special marking required in the contract or order, marking of packages and shipping containers shall be in accordance with MIL-STD-129.)

6. NOTES

6.1 Intended use. -

6.1.1 Type A heaters are intended for installation in machinery lubricating oil sump tanks to maintain the proper temperature and viscosity of the oil where little circulation or heat is provided (for example, reduction gear sumps).

6.1.2 Type B heaters are for use in pressure lubricating oil systems where heat is required in purifying the oil, or to raise the temperature to operating limits required for the applicable machinery.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Location of holes for cable connections to terminal box (see 3.2.3).
(d) Whether a magnetic contactor is required (see 3.2.6).
(e) Voltage required (see 3.2.8).
(f) Position of heaters if other than horizontal (see 3.3.1).
(g) Whether a safety switch or a liquid level control is required, and special setting, if necessary, when a safety switch is specified (see 3.3.2).
(h) Volume of oil to be heated, temperature rise and time (see 3.3.4).
(i) Whether insulation and lagging are required (see 3.4.5).
(j) Oil rate, gallons per hour (see 3.4.7).
(k) Number of manuals required (see 3.7).
(l) Preservation, packaging, packing and marking requirements if other than specified in 5.1 (see 5.2).

Preparing activity:
Navy - SH
(Project 4540-N0027)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4663, WHICH IS OBSOLETE

(NAVSHIPS OVPRT 12-66)

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